



Two-View Correspondence Pruning via Channel-Spatial Interaction and Bidirectional Consensus Interaction

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Abstract

Accurately identifying correct correspondences in two images is a crucial task in computer vision. Current methods predominantly use PointCN blocks as feature extraction backbones and learn local-global consensus through a progressive learning strategy. However, such methods have two main drawbacks: First, PointCN blocks, composed of multilayer perceptrons and normalization layers, process spatial positions independently, leading to limited interaction between channel-wise and spatial-wise dimensions. Second, the progressive learning strategy primarily focuses on unidirectional transfer from local to global consensus, yet neglects the bidirectional interaction between local and global consensus. To address these issues, we propose the Channel-Spatial interaction and Bidirectional Consensus interaction-Based Network (CSBCNet), which contains three innovative blocks: Channel-Spatial Interaction (CSI), Local Consensus Mining (LCM), and Global Consensus-Aware Attention (GCAA). Specifically, CSI enhances interaction between channel-wise and spatial-wise dimensions through a dual-path attention mechanism, addressing the limited interaction caused by the independent processing of spatial positions in PointCN blocks. LCM

extracts reliable local consensus by modeling geometric structures and spatial continuity within correspondences. GCAA captures global consensus by aggregating correspondences that are highly likely to be correct ones, and achieves bidirectional interaction between local and global consensus through cross attention. Experiments demonstrate our CSBCNet's superior performance in camera pose estimation and correspondence pruning. Notably, when the CSI block is applied to the existing OANet and MS2DGNet networks, it achieves significant performance improvements of 10.27% and 7.5%, respectively, on the mAP5° metric on the camera pose estimation task.

CCS Concepts

• **Computing methodologies** → **Matching**; • **Information systems** → **Similarity measures**.

Keywords

Correspondence pruning, feature matching, outlier removal, consensus learning, camera pose estimation.

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1 Introduction

In computer vision tasks, establishing accurate point-to-point correspondences between related images (i.e., feature matching) forms

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the foundation for tasks such as structure from motion (SfM) [3, 34], visual simultaneous localization and mapping (SLAM) [8], and image stitching [5]. Existing methods (*e.g.*, SIFT [21] and SuperPoint [6]) can be used to establish an initial correspondence set. However, due to the complexities in real-world scenarios, such as significant viewpoint changes, variations in lighting conditions, motion blur, or repetitive structures, the initial correspondence set usually contains a large number of incorrect matches (*i.e.*, outliers). Therefore, to ensure the accuracy of the matching results, it is essential to design effective methods to identify and remove these outliers while retaining as many correct correspondences (*i.e.*, inliers) as possible.

Outlier removal (*i.e.*, two-view correspondence pruning) methods can be broadly divided into two categories: traditional methods and learning-based methods. Among traditional methods, RANSAC [7] and its variants [25, 32] are the most widely used, primarily relying on sampling and verification strategies. Although these methods perform well in simple scenarios, their effectiveness significantly degrades as the outlier ratio increases. With the advancement of deep learning, learning-based outlier removal methods have emerged. The pioneering work, PointCN [36], redefines the outlier removal problem as a combination of inlier-outlier classification and essential matrix regression. It employs a Multilayer Perceptron (MLP) as its backbone, incorporating context normalization to capture implicit relationships between correspondences, and utilizes the weighted eight-point algorithm for camera pose estimation.

Building upon PointCN [36], CLNet [43] achieves significant performance improvements by learning the consistency among correspondences. Specifically, CLNet [43] introduces a progressive learning strategy to gradually learn local-global consensus. CLNet [43] first identifies the nearest neighbors for each correspondence in feature space and aggregates them to capture local consistency (*i.e.*, local consensus). Then, it connects all correspondences to construct a global graph and employs a Graph Convolutional Network (GCN) [10] to further model global consistency (*i.e.*, global consensus).

Building upon the progressive learning strategy, subsequent methods focus on designing diverse modules to better capture consensus information. Specifically, NCMNet [16] enhances local consensus capture by searching for nearest neighbors in three spaces: geometric, feature, and graph spaces, thereby providing more comprehensive representations. GCTNet [9] employs two aggregation methods, maxpooling and annular convolution [43], to extract local consensus and explores the interactions between different local consensus. VSFormer [11] captures both local and global consensus by fusing image and spatial information.

Although these methods have achieved performance improvements, several drawbacks remain: 1) Current methods primarily use PointCN blocks [36] as feature extraction backbones, which process correspondences spatial positions independently, restricting interaction between channel-wise and spatial-wise dimensions. This drawback limits the correspondence feature representations and further hurts the accuracy of correspondence pruning. 2) The existing methods typically adopt a progressive learning strategy that enables unidirectional transfer from local to global consensus through cascading operations. This strategy neglects the bidirectional interaction between local and global consensus [23, 28, 29], failing to integrate both into more cohesive feature representations.

To address these issues, we first propose a novel and effective network backbone called the Channel-Spatial Interaction (CSI) block. This block adopts a dual-path attention mechanism framework that integrates channel-wise adaptive attention with spatial attention to enhance feature representation, thereby avoiding information loss caused by processing spatial positions independently of the PointCN blocks. Specifically, in the channel-wise dimension, adaptive attention dynamically learns the correlation strength between channels, enhancing critical channels while suppressing redundant noise. In the spatial-wise dimension, spatial attention utilizes global average pooling and max pooling to generate spatial statistics, optimizing the spatial distribution of features. Finally, the block merges the outputs from the original input, channel-wise, and spatial attentions through multi-residual connections, thereby extracting information-rich feature representations. Serving as a new network backbone, the CSI block achieves significant performance improvements compared to the PointCN [36] blocks, maintaining reasonable computational costs as indicated in Table 6.

In addition, to explore the bidirectional interaction between local consensus and global consensus and generate more cohesive feature representations, we propose local consensus mining (LCM) block and global consensus Aware Attention (GCAA) block. On one hand, the LCM block captures explicit dynamic features through KNN graph construction [43], which models the geometric structures among correspondences. On the other hand, the LCM block extracts the sequence-aware features by the OABlock [37] to capture spatial continuity. By integrating these two types of features, the block reliably extracts local consensus by jointly modeling geometric structures and spatial continuity within correspondences, serving as the foundation for global consensus extraction. The GCAA block builds global consensus by aggregating correspondences that are highly likely to be correct ones, and captures interaction between local and global consensus using cross-attention mechanisms. Specifically, GCAA first filters high-confidence correspondences through linear projection to reduce interference from erroneous local consensus. It then captures global consensus using multi-vector tokens and enhances semantic understanding through self-attention. Finally, inverse cross-attention propagates global consensus back to all correspondences, achieving bidirectional interaction between local and global consensus. Based on the aforementioned modules, we have designed a novel and effective network called the Channel-Spatial interaction and Bidirectional Consensus interaction-Based Network (CSBCNet). The main contributions of this work can be summarized as follows:

- We propose the CSI block as a novel and effective backbone, which addresses the information loss problem in PointCN caused by the independent processing of spatial positions. By introducing a dual-path attention mechanism that combines channel-wise adaptive attention and spatial attention, it enhances the feature representation of correspondences.
- We design the LCM block, which achieves robust local consensus learning by fusing explicit dynamic features with sequence-aware features.
- We propose the GCAA block. This block builds global consensus by aggregating correspondences that are highly likely to be correct, and enables bidirectional interaction between

local and global consensus through cross-attention, addressing the neglect of the interaction in existing methods.

2 Related Work

2.1 Traditional Methods

Traditional methods such as RANSAC [7] have been commonly used for correspondence pruning. RANSAC [7] employs a hypothesis-and-verification framework, repeatedly selecting subsets of data to estimate the best model for identifying inliers. Many variants of RANSAC [7], such as NAPSAC [32], USAC [27], and MAGSAC [1] employ different strategies to improve the computational efficiency and effectiveness, achieving notable success. Specifically, NAPSAC [32] samples within local neighborhoods to enhance the capture of local structural consistency, thereby boosting robustness. USAC [27] optimizes hypothesis generation by integrating advanced sampling strategy and model verification technology, improving the efficiency and accuracy of RANSAC. MAGSAC [1], based on a probabilistic model, enhances model scoring by estimating the uncertainty of data points, leading to greater robustness. However, these methods are highly sensitive to outliers [42]. Consequently, their performance is limited when data have a high proportion of outliers, making them difficult to meet the needs of practical applications. Therefore, researchers have increasingly turned to deep learning-based methods to achieve better performance.

2.2 Learning-Based Methods

In recent years, many learning-based methods show promising performance on the correspondence pruning task. PointNet [26] is a groundbreaking work that processes irregular point cloud data using an MLP backbone. Subsequently, PointCN [36] builds upon the PointNet architecture, and redefines correspondence pruning as a combination of binary classification and regression tasks. In addition, PointCN [36] introduces contextual normalization operations within the MLP layers to capture more comprehensive global information. Since then, numerous learning-based methods have been proposed. For instance, OANet [37] enhances inlier identification by combining differentiable pooling and unpooling layers with an ordered perception filtering module. LMCNet [18] performs inlier selection through learning motion coherence. CLNet [43] improves inlier detection by progressively pruning correspondences. ACNe [30] designs an attention mechanism for feature normalization. NMNet [41] defines a new neighborhood detection strategy to identify inliers. PGFNet [15] designs a feature grouping residual attention module to capture the contextual information of correspondences. NCMNet [16] selects inliers by exploring domain consistency across three spaces: geometric, feature, and graph spaces. VSFormer [11] selects inliers by fusing image and spatial information. These learning-based methods employ different strategies and achieve significant performance improvements.

However, most of these methods predominantly utilize PointCN blocks as the feature extraction backbone. PointCN blocks process spatial positions independently, thus failing to fully exploit the potential for feature interactions between channel and spatial dimensions. In contrast, the proposed CSI block fuses channel attention and spatial attention to strengthen the response of important feature channels on the channel dimension and reconfigure the

feature distribution patterns on the spatial dimension, ultimately obtaining more discriminative feature representations.

2.3 Consensus of Correspondences

In feature matching, inliers follow geometric constraints or homographic transformations, exhibiting consensus under 2D rigid transformations, while outliers appear random [12, 17, 22, 24]. This principle has inspired various correspondence pruning methods. Traditional approaches like BF [13], CODE [12], GMS [2], and RoSe [19] leverage different consensus measures (global, separable, statistical, or rotation-invariant) to identify inliers. Learning-based methods like CLNet [43], NCMNet [16], ConvMatch [39], and DeMatch [38] further advance this by incorporating progressive learning, multi-space neighborhood consensus, or motion vector analysis. In this paper, we capture global consensus by aggregating correspondences that are highly likely to be correct, and enable bidirectional interaction between local and global consensus through cross-attention.

3 Method

3.1 Problem Formulation

Given a pair of images (I, I'), we first employ existing methods (such as SIFT [21] or SuperPoint [6]) to extract keypoints. Then, an initial correspondence set S is built using a nearest neighbor matching strategy:

$$S = [s_1, s_2, \dots, s_N] \in \mathbb{R}^{N \times 4}, s_i = (x_i, y_i, x'_i, y'_i), \quad (1)$$

where s_i represents the i -th initial correspondence containing two keypoints: (x_i, y_i) in image I and (x'_i, y'_i) in image I' , and the initial correspondences have been normalized by camera intrinsic.

Since, the initial correspondence set S typically contains a significant number of outliers. Therefore, we need to identify inliers as accurately as possible while removing outliers. We propose CSBCNet to achieve this goal, as shown in Fig. 1. Following CLNet [43], our CSBCNet uses two cascaded sequential pruning modules to process the input set S and obtain the corresponding output.

$$(S_1, o_1) = f_{\delta_1}(S), \quad (S_2, o_2) = f_{\delta_2}(S_1, o_1), \quad (2)$$

where $f_{\delta_1}(\cdot)$ and $f_{\delta_2}(\cdot)$ represent two sequential pruning modules with parameters δ_1 and δ_2 . $S_1 \in \mathbb{R}^{N_1 \times 4}$ and $S_2 \in \mathbb{R}^{N_2 \times 4}$ denote the subsets obtained after performing pruning modules ($N > N_1 > N_2$), and o_1 and o_2 denote their logit values. The logits in o_2 are transformed through a linear layer to generate weight vector ω_2 for the correspondence subset S_2 , which are then used to estimate the essential matrix $\hat{E}$:

$$\hat{E} = g(S_2, \omega_2), \quad p = V(S, \hat{E}), \quad (3)$$

where $g(\cdot)$ represents the weighted eight-point algorithm, and $V(\cdot)$ denotes full-size verification using $\hat{E}$ to recover potentially mispruned inliers from S . The final output p represents the inlier probabilities for all correspondences.

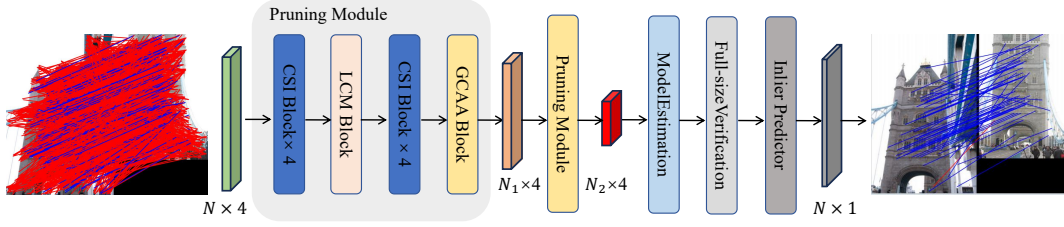


Figure 1: The architecture of our CSBCNet. $N \times 4$ represents the coordinates of N initial correspondences. Two pruning modules progressively reduce the input correspondences to a high-confidence candidate set of size N_2 . The essential matrix is then estimated using the candidate set. Finally, we perform full-size verification of the initial correspondences using the estimated essential matrix, obtaining an $N \times 1$ prediction probabilities for the initial correspondences.

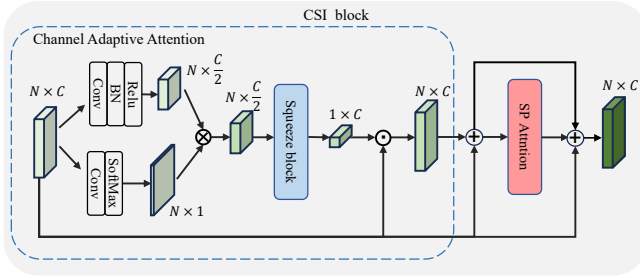


Figure 2: The architecture of the proposed CSI block. The CSI block mainly consists of two attention blocks: Channel Adaptive Attention (CA Attention) and Spatial Attention (SP Attention), as well as Multi-residual Connections.

3.2 The Network Framework

The overall architecture of CSBCNet is illustrated in Fig. 1, which mainly consists of two cascaded pruning modules, a model estimation module, and a full-size verification module. Each pruning module first maps the input correspondence set into a C -dimensional feature space via an MLP layer, followed by a multi-stage feature enhancement and consensus learning process: the first four CSI blocks perform initial feature enhancement, the LCM block extracts local geometric consensus, the next four CSI blocks further refine the feature representation, and finally, the GCAA block integrates global and local consensus information before a linear layer predicts correspondence weights. The correspondences filtered through the two-stage pruning process are fed into the model estimation module to compute the essential matrix. This estimated matrix is then passed to the full-size verification module, which performs a comprehensive evaluation to classify all initial correspondences in S .

3.3 Channel-Spatial Interaction Block

To address the insufficient interaction between channel and spatial dimensions caused by isotropic processing in PointCN blocks, we propose a Channel-Spatial Interaction (CSI) block, as shown in Fig. 2. This block contains three core components: 1) Channel-adaptive attention dynamically enhances crucial channels through learnable weights while suppressing noisy channels. 2) Spatial attention optimizes spatial distribution by combining dual-pooling operations

with lightweight MLPs, effectively suppressing outliers. 3) Multi-residual connections directly aggregate original, channel-enhanced, and spatial-enhanced features through hierarchical residual fusion, achieving multi-dimensional feature complementarity while preserving underlying geometric information.

1) **Channel Adaptive Attention.** The Channel Adaptive Attention is inspired by the Polarized Self-Attention [14], and its architectural details are illustrated in Fig. 2. We first obtain the query vector $\mathbf{W}_q \in \mathbb{R}^{N \times C/2}$ and key vector $\mathbf{W}_v \in \mathbb{R}^{N \times 1}$ by passing the input feature $\mathbf{F} \in \mathbb{R}^{N \times C}$ through different convolutional layers, where C represents channel dimensions:

$$\mathbf{W}_v = \text{ReLU}(\text{BN}(\text{Conv}_1(\mathbf{F}))), \mathbf{W}_q = \text{Conv}_2(\mathbf{F}), \quad (4)$$

where BN denotes the batch normalization operation, ReLU is the activation function and Conv_1 and Conv_2 represent convolution operations. Next, we compute the channel attention weight set $\mathbf{W}_z \in \mathbb{R}^{1 \times (C/2)}$ as follows:

$$\mathbf{W}_z = \mathbf{W}_q^T \otimes \text{SoftMax}(\mathbf{W}_v), \quad (5)$$

where $\otimes$ represents matrix multiplication. Finally, $\mathbf{W}_z$ is further refined to generate the final attention weights:

$$\mathbf{W}_{ca} = \text{Squeeze}(\mathbf{W}_z), \quad (6)$$

where *Squeeze* contains the following operations: the layer normalization, the sigmoid activation, the convolution, and the batch normalization. The final channel weights $\mathbf{W}_{ca} \in \mathbb{R}^{1 \times C}$ are applied to the original feature $\mathbf{F}$, resulting in the channel-enhanced feature $\mathbf{F}_{ch} \in \mathbb{R}^{N \times C}$.

$$\mathbf{F}_{ch} = \mathbf{F} \odot \mathbf{W}_{ca}, \quad (7)$$

where $\odot$ represents the element-wise multiplication.

2) **Spatial Attention.** To capture the significance of different spatial locations and further enhance the feature representation, we incorporate Spatial Attention [15].

The spatial attention block processes the channel-adaptive output feature map $\mathbf{F}_{ch}$ by first aggregating spatial information through parallel average pooling and max pooling. The summed results of the two pooling operations are fed into an MLP to generate spatial weight set, which are element-wise multiplied with $\mathbf{F}_{ch}$ to produce $\mathbf{F}_{sp}$. The process is formulated as:

$$\mathbf{F}_{sp} = \mathbf{F}_{ch} \odot \text{MLP}(\text{APool}(\mathbf{F}_{ch}) + \text{MPool}(\mathbf{F}_{ch})) + \mathbf{F}_{ch}, \quad (8)$$

where $\odot$ represents element-wise multiplication, and *APool* and *MPool* denote average pooling and max pooling, respectively.

3) **Multi-residual Connections.** The CSI block enables the model to preserve the original features while progressively integrating important spatial and channel information through the introduction of multi-residual connections. This enhances the richness and accuracy of feature representation. The specific formula for the multi-residual connections is as follows:

$$\mathbf{F}_{out} = \mathbf{F} + \mathbf{F}_{ch} + \mathbf{F}_{sp}, \quad (9)$$

where $\mathbf{F}$ represents the initial input, and $\mathbf{F}_{ch}$ ($\mathbf{F}_{sp}$) is the output of the channel adaptive attention (spatial attention) block.

3.4 Local Consensus Mining Block

To capture more reliable local consensus, we propose the Local Consensus Mining (LCM) block. This block integrates explicit dynamic graph features and sequence-aware features, efficiently extracting and representing the geometric structure and the spatial continuity of correspondences to capture more reliable local consensus.

1) **Feature Mining.** To capture local consensus, we construct an explicit dynamic graph $\mathbf{G}_{ex} \in \mathbb{R}^{N \times C \times K}$ based on $\mathbf{F}_{out} = \{f_1, f_2, \dots, f_N\}$. The explicit dynamic graph $\mathbf{G}_{ex}$ is defined as a set of local graphs $\mathbf{G}_i = \{\mathbf{V}_i, \mathbf{E}_i\}$, where $\mathbf{V}_i = \{f_{i1}, f_{i2}, \dots, f_{ik_i}\}$ represents the K nearest neighbors of f_i , and $\mathbf{E}_i = \{e_{i1}, e_{i2}, \dots, e_{ki}\}$ denotes the relationships between f_i and its neighbors. Each edge e_{ij} is constructed by concatenating f_i and its neighbor f_j :

$$e_{ij} = [f_i \| f_i - f_j], \quad (10)$$

where $[\cdot \| \cdot]$ denotes the concatenation operation. Finally, we aggregate the information using circular convolution to obtain the dynamic graph feature set $\mathbf{F}_{ex} = \text{Conv}_3(\text{Conv}_4(\mathbf{G}_{ex}))$, where Conv_3 and Conv_4 represent convolutional layers with $1 \times g$ and $1 \times k/g$ kernels, respectively.

When constructing the dynamic graph, we use OABlock [37] to capture the local spatial continuity of correspondences. The OABlock implicitly clusters the features into multiple clusters and captures the consensus information within each cluster through the OAFiltering operation [37], resulting in the sequence-aware feature set $\mathbf{F}_{im} = \text{OA}(\mathbf{F}_{out})$, where OA represents the OABlock.

2) **Feature Fusion.** To fully leverage the diverse information from different local features, it is essential to integrate them effectively: $\mathbf{F}_{local} = \text{MLP}([\mathbf{F}_{ex} \| \mathbf{F}_{im}])$.

3.5 Global Consensus Aware Attention Block

Although the LCM block effectively models local consensus information from multiple perspectives, relying solely on local consensus fails to capture the global structural relationships among inliers. In practical matching tasks, local consensus is susceptible to outliers, leading to inaccurate inlier-outlier discrimination and thereby undermining the effectiveness of correspondence pruning. Therefore, introducing global consensus is crucial for improving pruning precision. To this end, we propose the Global Consensus Aware Attention (GCAA) block, which aims to uncover latent structural consistency among correspondences from a global perspective, as shown in Fig. 3. Inspired by the use of the CLS token in Vision Transformers [20] for capturing global features, we introduce learnable tokens in the GCAA block to capture global consensus. Guided by local consensus, the block prioritizes high-confidence correspondences to construct global consensus and employs cross-attention

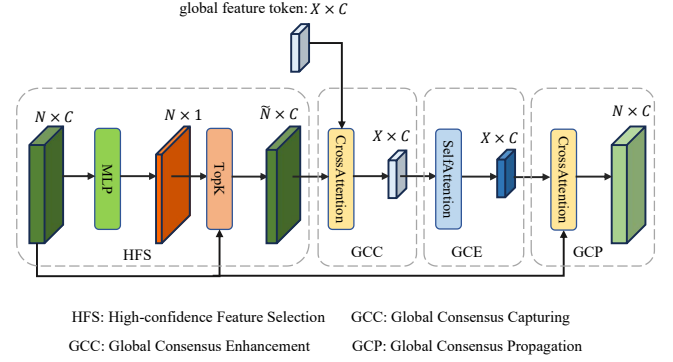


Figure 3: The architecture of the proposed Global Consensus-aware Attention Block. Global feature tokens are multiple learnable feature vectors.

mechanisms to propagate global consensus back to all correspondences, thereby enabling bidirectional interaction between local and global consensus. This block comprises four key steps: High-confidence feature selection, global consensus capturing, global consensus enhancement, and global consensus propagation.

1) **High-confidence Feature Selection.** First, the confidence score of each correspondence is calculated through the linear layer. To minimize the influence of a large number of outliers on the capture of global consensus, we select a subset of correspondences with higher confidence scores as the candidate source for global consensus capturing: $\mathbf{F}_{top} = \text{TopK}(\mathbf{F}_g, \text{Project}(\mathbf{F}_g), sp)$, where $\mathbf{F}_g \in \mathbb{R}^{N \times C}$ represents the input features $\mathbf{F}_{local}$ after processing by 4 CSI block. $\text{Project}(\cdot)$ refers to the operation that projects $\mathbf{F}_g$ into a one-dimensional space to obtain the probability distribution p for each correspondence, and $\text{TopK}(\cdot)$ selects the top sp proportion of features $\mathbf{F}_g$ based on the probability p .

2) **Global Consensus Capturing.** To capture global consensus, we introduce a set of learnable global feature tokens $\mathbf{F}_{global} \in \mathbb{R}^{X \times C}$, where X is the number of tokens and C denotes the feature dimension. These tokens are randomly initialized at the beginning of the network and iteratively refined during training to represent the local consensus among correspondences from a global perspective. After obtaining high-confidence features in the previous stage, we capture the global consensus through the cross attention operation [33]:

$$\mathbf{F}'_{global} = \text{CrossAttention}(\mathbf{F}_{global}, \mathbf{F}_{top}, \mathbf{F}_{top}), \quad (11)$$

3) **Global Consensus Enhancement.** To further enhance the global consensus, we apply a self attention [33] to enable the internal information of $\mathbf{F}'_{global} \in \mathbb{R}^{N \times C}$ to interact and capture long-range dependencies: $\mathbf{F}^e_{global} = \text{SelfAttention}(\mathbf{F}'_{global}, \mathbf{F}'_{global}, \mathbf{F}'_{global})$.

4) **Global Consensus Propagation.** After obtaining the global feature embedding, we interact the global consensus with the original input features, ensuring that the features of all correspondences retain local consensus information while also integrating global consensus information:

$$\mathbf{F}_b = \text{CrossAttention}(\mathbf{F}_g, \mathbf{F}^e_{global}, \mathbf{F}^e_{global}). \quad (12)$$

Table 1: Pose estimation performance comparisons on YFCC100M and SUN3D datasets. Bold indicates the best.

Method	YFCC100M (%)				SUN3D (%)			
	Known Scene		Unknown Scene		Known Scene		Unknown Scene	
	5°	20°	5°	20°	5°	20°	5°	20°
RANSAC[7]	5.74	16.67	9.08	22.80	4.43	15.38	2.86	11.22
PointCN [36]	14.51	35.82	23.71	50.57	11.93	36.03	9.73	33.09
OANet[37]	33.05	56.81	39.23	67.63	21.42	48.77	16.37	42.22
TNet[44]	40.86	63.81	46.74	73.11	23.55	50.99	17.69	44.03
CLNet [43]	39.25	63.01	52.45	75.97	20.30	45.73	17.26	41.32
MS ² DGNet[4]	37.78	62.78	46.98	75.13	22.93	50.67	17.34	43.41
PGFNet[15]	44.20	66.67	53.70	75.79	23.87	51.13	17.45	40.06
VSFormer[11]	48.80	70.14	62.68	80.88	24.34	50.63	20.07	45.36
GCTNet[9]	49.00	71.26	63.77	82.86	24.87	51.25	20.42	45.94
CSBCNet	55.75	75.24	68.18	84.58	25.47	51.94	21.58	46.78

Through the four steps mentioned above, we can obtain the features that integrate local and global consensus. The integration of the two consensus improves the discrimination between inliers and outliers.

3.6 Loss Function

Following CLNet [43] and OANet [37], we employ a hybrid loss function to supervise our network:

$$loss = l_c + \lambda l_{reg}(\mathbf{E}, \hat{\mathbf{E}}), \quad (13)$$

where λ is a parameter used to balance the two losses. l_c is the binary cross-entropy loss with self-adaptive temperature proposed by CLNet [43], and l_{reg} is the essential matrix loss calculated using Sampson distance. $\mathbf{E}$ and $\hat{\mathbf{E}}$ denote ground-truth and estimated essential matrices.

4 Experiments

4.1 Datasets

In our experiments, to validate the effectiveness of the proposed model, we employ both indoor and outdoor datasets. For the outdoor dataset, we select YFCC100M [31], which contains 100 million images depicting a wide range of real-world scenes and is divided into 72 sequences. Following [37], we select 4 sequences as unknown sequences, while the remaining 67 sequences are treated as known sequences. For the indoor dataset, we utilize SUN3D [35], which comprises 254 indoor video sequences. Among these, 239 sequences are designated as known sequences, and the remaining 15 sequences are treated as unknown sequences.

4.2 Implementation Details

Unless otherwise stated, the methods in the table all use SIFT to construct the initial correspondences following the experimental settings of [37]. The feature dimension C is set to 128. We implement iterative pruning using two sequential pruning modules. For the choice of neighbor count, we select $K = 9$ for the first module and $K = 6$ for the second module. In the configuration of annular convolutions, we choose the numbers of groups g to be 3 and 2 for

two modules, respectively. Each CSI block contains normalization layers before its channel and spatial attention blocks for stable optimization. These layers (MLP, InstanceNorm, and BatchNorm) enhance feature conditioning while preserving the core attention mechanism in Fig 2. Following [37], we use the Adam optimizer [40] with a fixed learning rate of 10^{-3} for network training. The training consists of 500k iterations. The weight parameter r for the mixed loss function is initialized to 0 and fixed at 0.5 after the first 20k iterations.

4.3 Evaluation Metrics

We evaluate our model on two primary tasks: Camera pose estimation and Correspondence pruning. For camera pose estimation, we adopt mean Average Precision (mAP) at 5° and 20° error thresholds to measure the accuracy of predicted poses relative to ground truth. For Correspondence Pruning, we use Precision, Recall, and F-score as evaluation metrics, where Precision is the ratio of correctly identified inliers among predicted correspondences. Recall is the ratio of correctly identified inliers among all ground-truth inliers, and the F-score is the harmonic mean of Precision and Recall.

4.4 Camera Pose Estimation

As shown in Table 1, our CSBCNet is compared with 9 correspondence pruning methods across two public datasets. These methods include both traditional and learning-based methods: RANSAC [7], PointCN [36], TNet [44], OANet [37], CLNet [43], MS²DGNet [4], PGFNet [15], VSFormer [11], and GCTNet [9].

The results in Table 1 show that learning-based methods outperform traditional methods. Among the learning-based methods, our proposed method achieves the best results in both indoor and outdoor scenes. Specifically, compared to the traditional method RANSAC, our method improves the mAP5° and mAP20° test results by 59.1% and 61.78%, respectively, in outdoor unknown scenes, and by 50.01% and 58.57%, respectively, in outdoor known scenes. In addition, compared to the current state-of-the-art learning-based methods, our method surpasses the second-best GCTNet by 6.75% and 3.98% in mAP5° and mAP20° in outdoor known scenes, and by 4.41% and 1.72% in unknown scenes. For the indoor dataset, our

Table 2: Comparisons of Pose estimation performance on unknown YFCC100M without/with RANSAC as a post-processing step.

Method	YFCC100M(%)		SUN3D(%)	
	-	RANSAC	-	RANSAC
RANSAC [36]	-	9.08	-	2.86
PointCN [36]	23.71	49.67	9.73	15.49
OANet [37]	39.23	53.10	16.37	17.44
CLNet [43]	52.45	58.93	17.26	19.56
VSFormer [11]	62.68	63.32	20.08	19.88
GCTNet [9]	63.77	62.70	20.42	20.25
CSBCNet	68.18	64.10	21.58	20.49

Table 3: Comparisons of Pose estimation performance using SIFT and SuperPoint on the Unknown YFCC100M.

Method	SIFT		SuperPoint	
	5°	20°	5°	20°
RANSAC [7]	9.08	22.80	17.47	38.83
PointCN [36]	23.71	50.57	24.25	52.70
OANet [37]	39.23	67.63	35.27	66.81
CLNet [43]	52.45	75.97	39.19	67.37
VSFormer[11]	62.68	80.88	45.65	73.49
GCTNet [9]	63.77	82.86	47.17	74.76
CSBCNet	68.18	84.58	51.70	76.94

Table 4: Correspondence pruning performance comparisons on YFCC100M datasets.

Method	Known Scene			Unknown Scene		
	P(%)	R(%)	F(%)	P(%)	R(%)	F(%)
RANSAC [7]	47.35	52.39	49.47	43.55	50.65	46.83
PointCN [36]	54.43	86.88	66.93	52.84	85.68	65.37
OANet [37]	58.51	90.10	70.95	54.54	86.64	66.94
T-Net [44]	61.76	89.62	73.12	57.63	86.57	69.19
CLNet [43]	75.92	79.46	77.65	74.94	76.71	75.81
MS ² DGNet [4]	62.31	90.37	73.75	58.67	87.67	70.28
PGFNet [15]	60.36	89.88	72.22	57.46	87.87	69.48
VSFormer [11]	78.36	82.41	80.33	76.94	79.21	78.05
GCTNet [9]	78.50	82.24	80.32	77.00	79.02	78.00
CSBCNet	79.35	83.09	81.18	77.79	79.76	78.76

method exceeds the second-best GCTNet by 1.16% and 0.84% in mAP5° and mAP20° in unknown scenes. These results fully validate the effectiveness of our proposed method by using channel-spatial interaction and bidirectional consensus interaction.

Table 2 presents the experimental results when RANSAC [7] is used as a post-processing step. Although our method still achieves the best results across all scenarios, it is worth noting that the overall performance of the network shows a slight decline. This phenomenon can be attributed to the fact that our method already provides highly accurate inlier weights. Consequently, RANSAC

does not use the weights to guide its data subset sampling process during the post-processing.

We further evaluate the impact of different feature extraction methods on the performance of our model when establishing initial correspondences. Specifically, we employ a learning-based method called SuperPoint [8], which uses a fully convolutional neural network to perform keypoint point detection in a self-supervised manner. We compare the results of traditional SIFT and the learning-based SuperPoint in outdoor unknown scenarios, as shown in Table 3. When initial correspondences are established using SuperPoint, our method continues to achieve the best results. Notably, initial correspondences established using SIFT outperform those established with SuperPoint in terms of pose estimation results. This is because SIFT is able to capture more precise keypoint locations than SuperPoint[16, 43]. These more precise keypoint locations make the better performance in our model.

4.5 Correspondence Pruning

As shown in Table 4, we compare our method against nine correspondence pruning methods. The results indicate that, in this task, deep learning-based methods significantly outperform the traditional algorithm RANSAC [7] across all three evaluation metrics. Notably, our CSBCNet surpasses other deep learning-based methods. Specifically, in both outdoor known and unknown scenes, our method increases the F-score by 0.85% and 0.71%, respectively, compared to the state-of-the-art method VSFormer [11].

It is worth noting that due to the different strategies employed by various methods in determining whether correspondences are inliers, there are significant discrepancies in their recall rates. Specifically, PointCN [36], OANet [37], MS²DGNet [4], PGFNet [15], and TNet [44] utilize a logit-based method to identify inliers, which enables a relatively high recall rate but results in lower precision. In contrast, CLNet [43], VSFormer [11], GCTNet [9], and our method employ an essential matrix-based strategy to predict inliers. Although these methods lead to a lower recall rate, they achieve higher precision. Consequently, a more reasonable assessment of the overall performance can be made by comparing the F scores.

To further showcase our method's effectiveness, we visualize the results of correspondence pruning, as shown in Fig. 4. From the results, it can be observed that our method performs particularly well in handling challenging scenarios such as large viewpoint changes (the first and second rows), repetitive structures (the third row), and weak textures (the fourth row).

4.6 Ablation Studies

1) **Main components.** To validate each block in CSBCNet, we incrementally add them to the baseline. Table 5 shows: "IPS" (iterative pruning with PointCN blocks), "CSI" (replacing PointCN blocks with CSI blocks), and adding "LCM" or "GCAA" separately. Results (Table 5) prove each component boosts performance. Notably, CSI improves accuracy by 15.12%, overcoming PointCN blocks' channel-spatial interaction limitation. Overall, CSBCNet achieves a 23.25% mAP5 gain over the baseline.

2) **Study on the Role of the CSI block.** We validate the CSI block by integrating it into OANet and MS²DGNet, comparing pose estimation performance. Table 6 shows significant gains: OANet

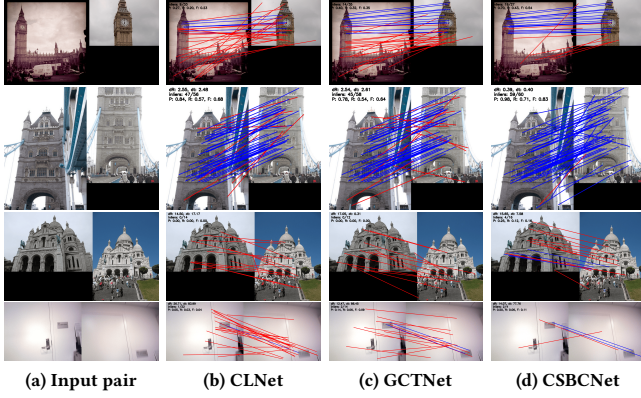


Figure 4: Visualization results of the correspondence pruning task to different methods. Inliers and outliers are depicted by red lines and blue lines, respectively.

Table 5: Ablation studies regarding performance gains of the key components.

IPS	CSI	LCM	GCAA	mAP5°	mAP20°
✓				44.93	73.60
✓	✓			60.05	79.39
✓	✓	✓		66.30	83.24
✓	✓	✓	✓	68.18	84.58

Table 6: Camera pose estimation results for networks with and without the CSI block (“+” represents using the CSI block)

Method	Known Scene		Unknown Scene		Size(M)
	5°	20°	5°	20°	
OANet	33.05	56.81	39.23	67.63	2.47
OANet ⁺	39.50	63.04	49.50	74.38	2.69
MS ² DGNet	37.78	62.78	46.98	75.13	2.61
MS ² DGNet ⁺	43.16	70.59	54.48	78.80	2.83

improves by 10.27% (mAP5, outdoor scenes), and MS²DGNet by 7.5%, proving CSI’s effectiveness and generalization.

3) Investigation of the GCAA block’s Function. To verify the effectiveness and transferability of the GCAA block, we conducted the following experiments: we replace the global consensus learning block in CLNet (which constructs a global graph and uses GCN to capture global consensus) with the GCAA block. The results are shown in the Table 7. The GCAA block achieves the best performance, demonstrating its effectiveness in capturing global consensus and highlighting the importance of capturing the bidirectional interaction between local and global consensus.

4) The Value of X and sp in GCAA. Ablation studies (Table 8) on YFCC100M reveal optimal GCAA parameters: $X = 48$ for global features and $sp = 0.2$ for correspondence selection. Larger

Table 7: Comparisons of pose estimation effects using Different global consensus capture methods. L: Local consensus learning only; G: Using GCN; W: Using GCAA block.

Method	mAP5°	mAP20°
CLNet-L	49.20	74.80
CLNet-G	52.45	75.97
CLNet-W	54.40	77.58

Table 8: Analysis of the value of X and sp in GCAA on the YFCC100M dataset, showing mAP5°, mAP10°, and mAP20° for different settings in outdoor unknown scenes.

Metric	X			sp		
	24	48	96	0.1	0.2	0.5
mAP5°	67.05	68.18	67.08	67.23	68.18	67.53
mAP20°	84.28	84.58	84.31	84.30	84.58	84.42

Table 9: Comparisons of computational resource consumption between our method and other state-of-the-art methods. SIZE: Model parameter size; F: Number of floating-point operations; Memory: Peak memory usage.

Methods	Size(M)↓	F(G)↓	Memory(M)↓	mAP5°(%)↑
VSFormer [11]	2.27	7.61	238.09	62.68
GCTNet [9]	4.09	10.33	126.91	63.77
CSBCNet	4.09	4.44	75.03	68.18

X causes redundancy while smaller X limits global consensus capture. Similarly, higher sp introduces outliers while lower sp retains insufficient inliers. These results highlight the need for balanced parameters to optimize global-local feature interaction.

5) Computational Analysis. Table 9 compares the computational complexity of our method with state-of-the-art methods, where arrow directions indicate metric optimization trends. Our method achieves the best performance with efficient computational costs, notably requiring fewer FLOPs and lower peak memory usage.

5 Conclusion

This paper presents CSBCNet, a correspondence pruning network featuring three core innovations: 1) CSI block employs dual-path attention to enhance channel-spatial feature interaction, addressing isotropic feature limitations in the PointCN block. 2) LCM block extracts local consensus through explicit graph construction and multi-perspective semantic clustering. 3) GCAA block enables local-global consensus via confidence filtering and cross-attention. Experimental results show that our CSBCNet achieves state-of-the-art performance in pose estimation and correspondence pruning.

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